

Q1444

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(Known) $m_A, m_B,$ E_p, v_0, θ

$$\vec{v}_A = \vec{v}_0 + \vec{v}_A'$$

(Vector)

$$\vec{v}_B = \vec{v}_0 + \vec{v}_B'$$

$$(Cons. Momentum) (m_A v_A)_x + (m_B v_B)_x = (m_A + m_B) v_0 \quad (1)$$

$$(Projecting) \begin{cases} v_{Ax} = v_0 - v_A' \cos 30^\circ \\ v_{Bx} = v_0 + v_B' \cos 30^\circ \end{cases}$$

$$m_A (v_0 - v_A' \cos 30^\circ) + m_B (v_0 + v_B' \cos 30^\circ) = (m_A + m_B) v_0 \quad (2)$$

$$m_A v_A' = m_B v_B', \quad v_B' = \frac{5}{3} v_A'$$

$$E_p + \frac{1}{2}(m_A + m_B) v_0^2 = \frac{1}{2} m_A v_A'^2 + \frac{1}{2} m_B v_B'^2 \quad (3)$$

$$(Geometry) \quad v_A^2 = v_0^2 + v_A'^2 - 2 v_0 v_A' \cos 30^\circ$$

$$Relativistic \quad v_B^2 = v_0^2 + v_B'^2 - 2 v_0 v_B' \cos 30^\circ$$

$$\cancel{40A} \quad v_A' = v_B', \quad E_p$$

$$14 \quad v_A', v_B', v_B' = \frac{5}{3} v_A', E_p = 120 \text{ J}$$

$$120 = 2.333 v_A'^2, \quad \therefore v_A' = 6 \text{ m/s}$$

$$\therefore v_B' = 10 \text{ m/s}$$

$$v_A = 4.11 \text{ m/s}$$

$$v_B = 17.39 \text{ m/s}$$

Q 14.43

$$m_A v_A + m_B v_{Bx} = 0 \quad \text{--- (1)}$$

$$v_{Bx} = -\frac{m_A}{m_B} v_A$$

$$\cancel{\frac{1}{2}(m_A + m_B)} h = l(1 - \cos \theta)$$

$$m_B g l (1 - \cos \theta) = \frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B v_B^2 \quad \text{--- (2)}$$

B directly under A ($v_{By} = 0$).

$$v_{Bx} = v_B = -\left(\frac{60}{40}\right) v_A$$

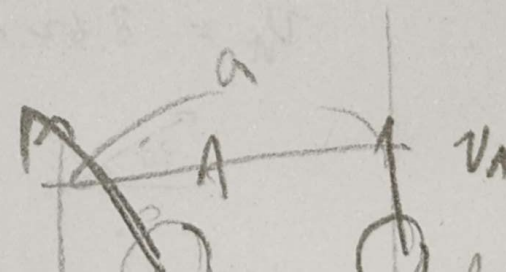
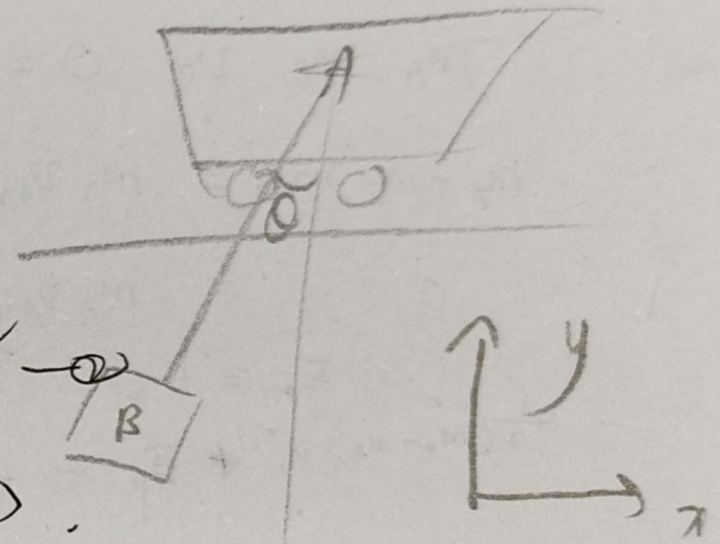
$$= -1.5 v_A$$

$$g = 9.8 \text{ m/s}^2 \Rightarrow 32.152 \text{ ft/s}^2$$

$$(40)(6)(1 - \cos 25^\circ) = \frac{1}{2} \left(\frac{60}{32.152}\right) (v_A^2) + \frac{1}{2} \left(\frac{40}{32.152}\right) \left(v_A^2 \frac{9}{4}\right)$$

$$v_A \approx -3.16 \text{ ft/s}$$

$$v_B \approx 4.66 \text{ ft/s}$$



$\frac{v}{A/G} \quad \angle \quad v_0$

ED

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~~$$(m \times vr)_A + (m \times vr)_B = (m \times vr)_A + (m \times vr)$$~~

Th

$\sqrt{A} \rightarrow (P)$

319°

$a = 2.73 \text{ m}$

4 Var $\left\{ \begin{array}{l} v_A, \\ v_B, \\ 0, \\ b, \end{array} \right.$

$$v_A = v_{oy} + v_{Agy}$$

$$\therefore V_{ox} = -V_{A/G(ox)}$$